Firm Size Distribution and Power Laws in Brazil: Empirical Evidence

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Abstract

This paper investigates power laws for the firm size distribution in Brazil for the largest 1000, 500 and 100 firms. A systematic nested testing strategy is implemented and stronger evidence for a Pareto type I distribution emerges only for the extreme upper tail of the distribution (top 100 firms). The Pareto exponents for the top 100 firms display a statistically significant downward trend and a structural break in 2010. Their mean falls from approximately 1.27 in 1999–2009 to 1.05 in 2010–2024, approaching the Zipf benchmark of 1 and indicating a thicker upper tail and greater revenue concentration among superstar firms.

Keywords: power law; firm size; Pareto distribution; Brazil

JEL Classification: C12; C49; L11

1. Introduction

The search for distributional regularities is valuable as an initial characterization of phenomena in different areas of knowledge. An influential example in economics relates to lognormal limiting distributions for firm size associated with the so-called Gibrat's law as implied by rates of firm growth for firms that are size-independent [see Sutton (1997) and Coad (2009)]. Quite often, regularities take the form of a power law, $Y = c\, X^{b}$, where Y and X are the variables of interest, c is a scaling constant and b is the exponent. The relation is ascribed for the upper tail of the distribution, and it is remarkable that such regularities appear to prevail in different contexts as in city sizes, firm sizes and stock market movements [see Gabaix (2009, 2016) for overviews]. The major challenge is to move further and uncover the underlying generating mechanisms that may eventually be attributed to well delineated economic processes that may involve possible evolutionary dynamics.

The present paper focuses on different aspects of the firm size distribution in Brazil. A handful of studies on Gibrat's law or its distributional implications in that context were undertaken by Resende (2004), Esteves (2007), Ribeiro (2007) and Resende and Cardoso (2013). The evidence strongly rejects the prevalence of Gibrat's law, except for some favorable results for the upper tail of the firm size distribution in one study. However, there is still potential for additional investigations on topics related to the firm size distribution that include power law features. In any case, dynamic aspects involving distributional changes and possible explanatory factors are likely to be relevant, especially in complex and heterogeneous economies. In fact, the relevance of possible long-run distributional shifts is discussed by Cabral and Mata (2003) for the case of a smaller economy like Portugal. The firm size distribution evolves towards a lognormal pattern and the calibration of a theoretical model, that incorporates financial constraints, displayed empirical consistency. Angelini and Generale (2008) further explored the role of financial constraints for the firm size distribution in Italy. Lotti and Santarelli (2004) also investigate firm size distributions in the Italian manufacturing industry. Newly created firms were followed on a quarterly base for six years and nonparametric methods were applied to different sectors. Altogether, the evidence that emerges from the aforementioned studies mostly indicates right skewed firm size distributions. Even though, one observes different convergence patterns of the distributions over time in the latter study while in principle one observes some consistency with the mechanisms addressed in the literature on firm selection and industry evolution appears to prevail. Luttmer (2007) considered the underlying firm-specific heterogeneities and the related implementation with

calibration methods favored a limiting firm size distribution that is a particular case of a Pareto distribution, namely the Zipf distribution.[1]

As for dynamic power law aspects it is worth mentioning Fernholz (2017), that proposes nonparametric econometric methods that characterize general power law distributions under basic stability conditions. Although the necessary requirements are not always met in all settings, it indicates that analyses that consider potentially changing distributions that are prone to a power law pattern may be fruitfully conducted.

Even when power law patterns are detected, the next natural step involves the discussion of possible generating mechanisms. Kumamoto and Kamihigashi (2018) attempt to map the possible classes of stochastic processes that can be consistent with power law patterns. Gabaix (2009, 2016) highlights underlying mechanisms associated with random growth, performance aspects related to extreme upper tails of the distribution ("economics of superstars") and transfer of power laws through aggregation.[2]

The study of power laws for firm size distribution still warrants further assessments. In the case of Brazil, Da Silva et al. (2018) addressed the prevalence of power laws for firm size, as measured by net revenue shares, in the case of the largest 1000 firms in Brazil in 2015. For that purpose, Pareto and Gumbel exponents were obtained upon the estimation of these assumed laws and the evidence is claimed to properly fit a Pareto law that is roughly consistent with the particular case of a Zipf law.[3] However, the upfront assumption of a simple power law process is questionable, although one should be less concerned on biases relating to OLS estimations given the large sample size in that particular application.[4] Furthermore, the authors emphasize the role of large firms in the Brazilian economy by which microeconomic shocks could have significant macroeconomic implications.[5] In fact as power laws relate to the upper portions of the size distribution, the so-called granular aspect may have important policy implications [see Rossi-Hansberg and Wright (2007) and Gabaix (2011)].

Finally, it is conceivable that distributional characteristics for firm size may change over a longer time horizon. For example, Rodrik (2016) contends that developing economies could be subject to premature deindustrialization. A related aspect may be associated with the reduction of the diversification of the economy. Hutchinson et al. (2010) find that in the U.K.

[1] It was initially conceived by Zipf (1949). An overview can be found in Saichev et al. (2010).

[2] Gabaix (1999) provides an attempt of uncovering generating mechanisms for the Zipf law in the case of city sizes.

[3] In many empirical applications, the OLS-based procedure could lead to significant biases in small samples as indicated by Gabaix and Ibragimov (2011), who suggested a possible correction. Furthermore, Urzúa (2011) had stressed, in the context of the particular case of the Pareto's law given by the Zipf´s law, that the referred procedure would be inappropriate, as the intercept is not a nuisance parameter in the regression. Schluter (2021) further elaborates on the estimation bias arising from OLS-based estimations for the Zipf law.

[4] Coad (2010) emphasizes the importance of not assuming more restrictive hypotheses when testing for the Pareto law.

[5] The oil firm Petrobrás is by far the largest in Brazil and operates in an extremely volatile sector.

and Belgium 4-digit industries, inter-industry diversification leads to a shift of the firm size distribution to the right with a stronger effect observed in the case of older age groups. Conversely, the possibility of a less diversified industry in Brazil, for example, could hypothetically lead to changes in power law patterns, if those are relevant. In fact, the relative loss of importance of specific sectors can potentially not only change firm size distributions but possibly the shape parameter of a power law, which typically holds a negative relation with firm size inequality. Such conjecture needs to be empirically addressed.

The present paper aims at contributing to the literature in aspects:

a) Systematic nested testing framework: by not assuming a simpler distribution (e.g., Pareto type I or Zipf) from the start, but rather nesting these models within more general models. In addition to Goerlich's (2013) LMp and Urzúa's (2000) LMz tests, we implement the PWL test of Urzúa (2020) which nests the Pareto type I model in the flexible Pareto type IV distribution;
b) Consideration of a large and heterogeneous emerging economy over a long time span (1999–2024). Brazil is likely to be prone to heterogeneous shocks of different natures and larger firms may benefit from spatial and sector-level advantages that are unavailable for smaller firms;
c) Structural break analysis: by conducting a time-series analysis of the tail exponents, which identifies a statistically significant downward structural break in 2010, coinciding with the commodity-boom period and signaling rising revenue concentration among superstar firms.

The remainder of the paper is organized as follows. The second section discusses basic conceptual aspects related to power law testing. The third section discusses the data and the obtained empirical results for the particular empirical application. The fourth section concludes.

2. Power laws and firm size distributions: empirical aspects

2.1 – Basic conceptual aspects

An influential power law emerges in the literature pertaining to the Pareto distribution that can be traced back to Pareto (1897) in the context of the assessment of income distribution. Useful overviews can be found in Johnson and Kotz (1970, Chap. 19). Specifically, the so-called Pareto type I distribution relates to a strictly positive continuous random variable $x \in \Re_{++}$ with probability density function (pdf) given by:

$$f_p(x) = \frac{\alpha}{\mu}\left[\frac{\mu}{x}\right]^{\alpha+1} \qquad x \geq \mu > 0,\ \alpha > 0 \qquad (1)$$

where μ stands for the location parameter and corresponds to the minimum value of x that indicates where the so-called "heavy tail" starts and can be intuitively, though not directly, associated with concepts of minimum scale of operation, whereas α indicates the shape

parameter that constitutes the main parameter of interest and is often known as the Pareto exponent and indicates how concentrated firm size is among very large firms. A salient feature relates to dispersion measures holding an inverse relation with the shape parameter α. In particular, the coefficient of variation is given by $CV(X)=1/\sqrt{\alpha(\alpha-2)}$ for α > 2, where the sole dependence on the shape parameter reflects a relative character.

A fruitful research strategy for testing power laws is to consider more general probability distributions that nest the Pareto type I. Goerlich (2013) suggests the consideration of a more general case given by a member of the Burr (1942) family of distributions [see Johnson and Kotz (1970), chap. 12.4.5 for an overview], also known as a Pareto type II distribution [see Arnold (2015)]. That more general distribution is characterized by the following probability density function (pdf) given by:

$$f_B=\frac{\alpha}{\sigma}\left(1+\frac{x-\mu}{\sigma}\right)^{-(\alpha+1)} \qquad x\geq\mu>0,\ \alpha>0,\ \sigma>0 \qquad (2)$$

In contrast with the simpler Pareto type I distribution, one introduces a location parameter μ and a scale parameter σ > 0, so that the lower bound is set by μ rather than being tied to the scale. In economic terms, σ controls the scale of the distribution independently of the degree of the heaviness of the upper tail.

Equation (2) nests Pareto type I when σ = μ and Zipf's law when σ = μ and α = 1. For α > 2, the variance is $Var(X)=\frac{\sigma^2\alpha}{(\alpha-1)^2(\alpha-2)}$, and the coefficient of variation is:

$$CV(X)=\frac{\sigma}{\sigma+(\alpha-1)\mu}\sqrt{\frac{\alpha}{\alpha-2}},\alpha>2.$$

Thus, for a fixed positive μ, the coefficient of variation depends on σ as well as α. The expression $\sqrt{\frac{\alpha}{\alpha-2}}$ applies only in the zero-location special case μ = 0.

Goerlich (2013) proposes to test the null hypothesis of H_0: $\sigma=\mu$ in (2) by means of a Lagrange multiplier (LM) test that requires only the maximum likelihood estimation of the restricted model, the Pareto type I case. Thus proceeding, one can obtain the following expression:

$$LM_P=n\frac{(\tilde{\alpha}+2)(\tilde{\alpha}+1)^4}{\tilde{\alpha}}\tilde{z}^2\sim_{asy}\ \chi^2(1)\ under\ \ H_0:\sigma=\mu \qquad (3)$$

where $\tilde{z}=\frac{\tilde{\alpha}}{\tilde{\alpha}+1}-\frac{1}{n}\sum_{i=1}^{n}\frac{\mu}{x_i}$ with $\tilde{\sigma}=\mu$ and $\tilde{\alpha}=\hat{\alpha}$ denoting restricted estimates. The computation of the test statistic is simple given the maximum likelihood estimates of $\hat{\alpha}$ obtained for the Pareto type I model. The author presents evidence that indicates satisfactory results from Monte Carlo simulations with good small sample properties and adequate power against some plausible alternatives.

Previously, Urzúa (2000) had advanced an analogous testing strategy for the Zipf law that considers the more specific joint null hypothesis of H_0: $\sigma = \mu$, $\alpha = 1$. Accordingly, it is possible to obtain the Lagrange multiplier (LM) test statistic that is given as follows:

$$LM_Z = 4n[z_1^2 + 6z_1z_2 + 12z_2^2] \sim_{asy} \chi^2(2) \quad under \quad H_0: \sigma = \mu \ , \alpha = 1 \qquad (4)$$

where $z_1 \equiv 1 - \frac{1}{n}\sum_{i=1}^{n} \ln\left(\frac{x_i}{\mu}\right)$ and $z_2 \equiv \frac{1}{2} - \frac{1}{n}\sum_{i=1}^{n} \frac{\mu}{x_i}$

Note that now one has an additional restriction under the null hypothesis in contrast with (3) and therefore a distinct number of degrees of freedom in the asymptotic distribution. Furthermore, the reported finite sample properties for the LMz test as obtained from Monte Carlo simulations appear as adequate. The particular interest in the law advanced by Zipf (1949) would lead, if tenable, to a strong inverse proportionality between firm rank and firm size. It indicates that the largest firm in an economy is approximately twice the size of the second largest, three times the size of the third largest, and so on.

The nested hypotheses motivate a sequential interpretation: LMp first tests Pareto type I within type II, and LMz then examines the additional Zipf restriction. In the empirical analysis, however, both tests are reported for every year and cutoff, including when LMp rejects. LMz tests the joint restrictions $\sigma = \mu$ and $\alpha = 1$ and serves as a separate diagnostic; its non-rejection following an LMp rejection does not establish Zipf's law.

2.2 – <u>More general nesting structures</u>

Urzúa (2020) argues that the contrasts between the null and alternative hypotheses would be subtle under the testing strategies advanced by Urzúa (2000) and Goerlich (2013) as the involved distributions would have a similar heavy-tail behavior. Thus, he suggests embedding the Pareto model within a more general structure of a Pareto type IV model. The more general Pareto type IV density can be expressed as follows [see Arnold (2015)]:

$$f(x) = \frac{\alpha}{\gamma\sigma}\left[1 + \left(\frac{x-\mu}{\sigma}\right)^{\frac{1}{\gamma}}\right]^{-(\alpha+1)}\left(\frac{x-\mu}{\sigma}\right)^{\frac{1}{\gamma}-1} \qquad for \quad x > \mu \qquad (5)$$

One can note an additional inequality parameter $\gamma > 0$. This parameter reflects the curvature of the distribution and captures inequality and, together with the shape parameter α, determines the tail of the distribution. The magnitude of that parameter indicates the local degree of asymmetry and the growth speed of transition between smaller and larger firms.

The imposition of the restrictions $\sigma = \mu$ and $\gamma = 1$ would readily lead to the Pareto type I model as given by expression (1). Additionally, if $\alpha = 1$, one would have the stricter case consistent with a Zipf law.

The author undertakes a testing procedure analogous to those indicated by the aforementioned works but with the more general alternative provided by a Pareto IV model. It can be shown that the corresponding Lagrange multiplier test is given as follows:

$$PWL = \hat{d}' \hat{I}^{-1} \hat{d} \quad \sim_{asy} \quad \chi^2(2) \quad under \;\; H_0 : \sigma = \mu \;\; , \gamma = 1 \qquad (6)$$

Where $\hat{d}$ = (d1,d2,0)′ with

$$d_1 = \frac{n\hat{\alpha}}{\mu} - (\hat{\alpha}+1)\sum_{i=1}^{n} \frac{1}{x_i} \qquad (7)$$

$$d_2 = -n + \hat{\alpha} \sum_{i=1}^{n} \ln\left(\frac{x_i - \mu}{\mu}\right) - (\hat{\alpha}+1)\sum_{i=1}^{n} \frac{\mu}{x_i} \ln\left(\frac{x_i - \mu}{\mu}\right) \qquad (8)$$

Here $\hat{\alpha}$ denotes the restricted Pareto type I shape estimate, and $\hat{I}^{-1}$ denotes the inverse of the information matrix.[6]

The reported Pareto type I maximum-likelihood (Hill, 1975) and Gabaix–Ibragimov (2011) rank-size estimates, and the LMp, LMz and PWL score statistics, were calculated in Python using NumPy and SciPy. LMp and LMz use the smallest included revenue as the cutoff in Tables 1–3; Table 6 uses the same convention for the Hill estimates. The top-100 PWL test uses the revenue of rank 101 as its cutoff, so every tested observation lies strictly above it. Its bootstrap p-values in Table 5 use 19,999 replications per year. The Wald test of the Zipf restriction, Mann–Kendall test, Sen's slope and weighted sup-Wald test were calculated in Python; the Wald and sup-Wald bootstraps use 20,000 replications. The Stata modules paretofit [Jenkins and Van Kerm (2015)] and pwlaw [Urzúa (2020)] are software references for Type I estimation and the PWL method, respectively; they are not called by the Python routines that reproduce the reported results.

The empirical strategy examines Pareto type I restrictions within Pareto types II and IV using LMp and PWL, respectively, together with the joint Zipf restrictions within type II using LMz. Pareto type III is not separately estimated or tested. The PWL comparison follows Urzúa (2020). This strategy provides a systematic assessment of power laws, but the two basic generating mechanisms discussed by Gabaix (2009, 2016) do not readily identify the underlying economic primitives. Those works emphasize:

(a) Random growth models with independent shocks contend that to achieve convergence to a steady-state distribution of firm size that is not lognormal but consistent with a power law, some friction is needed but still refers to a starting point that assumes Gibrat's law where firm growth is size-independent

(b) Transfer of power law from one variable to another by matching and optimization.

[6] Urzúa (2020) builds on Brazauskas (2002) for obtaining that component for the test statistic. The reader is referred to that work as the matrix is omitted for conciseness in the present paper.

It appears that there is still a gap of a better understanding of the economic factors that explain power laws. In any case, the importance of heterogeneous shocks may provide an additional motivation for studying the large and heterogeneous economy of Brazil. The present paper may constitute an intermediate step towards causal analyses in the future.

3. Empirical Analysis

3.1 – Data

This study makes use of balance sheet data for the 1000 largest firms in Brazil on an annual basis for the period 1999–2024. The coverage of the considered data sources is broader than more usual databases that focus only on publicly traded firms. In fact, the pioneering databases were constructed in connection with the periodical Conjuntura Econômica [Instituto Brasileiro de Economia, Fundação Getulio Vargas – IBRE-FGV] in terms of an annual special issue for the largest 500 firms and later a database for the 1000 largest firms was established while similar efforts were initiated by newspaper Valor Econômico. We were able to obtain data for 1999 and 2000 from IBRE-FGV and for 2001–2024 from Valor Econômico for the 1000 largest firms.

In the present study, the variable of interest will be the net revenue share of firm i in year (rs_{it}) computed upon the share of net revenue (NR) relative to the year's total value, which is readily defined for the n largest firms:

$$rs_{it} = \frac{NR_{it}}{\sum_{i=1}^{n} NR_{it}} \qquad (9)$$

The power law analyses will be implemented on a year-by-year basis for the referred upper tail of the distribution of net revenue shares and the top firms are not always the same in different years. The analysis also relates to Segarra and Teruel (2012), who examine firm-size distributions and the sensitivity of their characterization to sample size. Even though their focus is distinct from the present investigation, our findings highlight the importance of examining the upper tail separately with different cutoff points of the firm size distribution. As we shall see, the evidence of a Pareto type I distribution is strongest among Brazil's 100 largest firms, whose Pareto exponent exhibits a declining trajectory and a structural break in 2010.

3.2 – Empirical results

The empirical strategy tests exact Pareto type I restrictions within more general Pareto families. Table 1 reports the LMp and LMz score tests proposed by Goerlich (2013) and Urzúa

(2000). The statistics and their asymptotic probabilities are computed by the Python routines lmp_test and lmz_test, using the sample-minimum cutoff. Tables 1–3 retain the 999-replication bootstrap calibration from the historical Python analysis; the fiscal-2019 top-1000 statistics in Table 1 incorporate the restored source observation. Table 5 reports the separately implemented PWL test of Urzúa (2020), with bootstrap probabilities recalibrated in the V12 replication package.

INSERT TABLE 1 AROUND HERE

For the largest 1000 firms, LMp rejects Pareto type I against type II in every year of 1999–2024. LMz tests the joint restrictions $\sigma = \mu$ and $\alpha = 1$, corresponding to Zipf's law within type II. Its results are reported for completeness in every year, including when LMp rejects. Because the tests have different score statistics and critical values, their rejection decisions need not be ordered. We report both asymptotic and parametric bootstrap p-values.

Thus, for the largest 1000 firms in Brazil, the evidence is not consistent with a simple power law pattern. However, power laws typically prevail for the upper tail of some distributions of variables of interest. A further investigation possibility could consider a higher cutoff point for the Pareto distribution. In fact, in the context of firm size, another possibility is to focus on the 500 largest firms. The corresponding results for the tests are reported in Table 2.

INSERT TABLE 2 AROUND HERE

Under this stricter criterion, the evidence is still strongly against the prevalence of a power law. One cannot reject the null hypothesis for the 2012–2015 period, as suggested by the LMp test, and the evidence rejects a Pareto type I law in most of the years. Additionally, evidence from the LMz test favors the prevalence of a Zipf law for that upper tail of the firm size distribution in Brazil for that same period.

Finally, in Table 3, we consider a higher cutoff point in terms of the 100 largest firms in Brazil. In such case, a simple power law pattern, with a Pareto type I law, emerges in most years with exceptions occurring in the 2016–2020 period and in the 2022–2023 period. In fact, in the great majority of years the evidence is consistent with a Pareto type I and also with the particular case of Zipf law (with $\alpha = 1$), with the violation of the latter stricter case occurring in 2000, 2018, 2019 and 2022. However, in some years, when evidence suggests the rejection of a Pareto type I distribution driven by a single shape parameter, one cannot reject the stricter particular case of the Zipf law, as indicated by the LMz test; this is possible because the additional restriction implied by the structure of the test, together with sample variability, may place the statistic outside the rejection region, and the smaller sample size may also reduce the power of the test.

INSERT TABLE 3 AROUND HERE

Table 4 summarizes the same comparison across cutoffs. The exact Pareto type I restriction is rejected in at most ten of the 26 years for cutoffs up to the largest 300 firms (two years at n = 50, nine at n = 100 and two at n = 200), but in 23, 26 and 26 of 26 years at n = 500, 750 and 999, respectively, and the mean exponent falls from 1.311 at n = 50 to 0.823 at n = 999. The deterioration is therefore not monotonic: exact Pareto type I describes a narrow slice of the upper tail, roughly the 50 to 300 largest firms, and fails once the sample extends to half of the size distribution. These rejection counts are descriptive across a prespecified grid rather than selection-adjusted test decisions.

INSERT TABLE 4 AROUND HERE

Altogether, we must be cautious, but there is suggestive evidence for a heavier tail of the firm size distribution in terms of a Pareto type I pattern. However, it is important to consider a more general starting point with a more flexible Pareto type IV distribution. In Table 5, we report the results from the PWL Lagrange multiplier test by Urzúa (2020) that has as the null hypothesis the prevalence of a Pareto type I distribution.

INSERT TABLE 5 AROUND HERE

Table 5 generally does not reject exact Pareto type I, but the calibrated p-values reject in 2005, 2016, 2018, 2019, and 2022. Results around 0.05 are sensitive to Monte Carlo resolution, which motivates the 19,999-replication calibration. We do not equate non-rejection with evidence that no alternative fits as well.

INSERT TABLE 6 AROUND HERE

Having previously made the case for a Pareto type I distribution, Table 6 reports the sample-minimum Hill and Gabaix–Ibragimov rank-size estimates, with explicitly identified p-values for the Zipf restriction $\alpha = 1$. The MLE and OLS-GI columns each report asymptotic and parametric bootstrap probabilities.

To assess the temporal pattern of the annual top-100 Hill estimates, we consider the Mann–Kendall nonparametric test for a monotonic trend [see Mann (1945), Kendall (1975)]. Here T = 26 denotes the number of annual observations; n denotes the number of firms in an annual distributional test.

INSERT FIGURE 1 AROUND HERE

$$S=\sum_{i=1}^{T-1}\sum_{j=i+1}^{T} sgn\ (\hat{\alpha}_j-\hat{\alpha}_i) \qquad (10)$$

where x stands for the difference in parentheses, with

$$sgn\ (x)=\begin{cases} 1, & x>0, \\ 0, & x=0, \\ -1, & x<0. \end{cases}$$

The three cases indicate positive, zero and negative pairwise differences, respectively; the trend assessment depends on their aggregate S.

For a sufficiently large T, the asymptotic variance under H_0 is:

$$Var(S)=\frac{T(T-1)(2T+5)-\sum_{g=1}^{G} t_g(t_g-1)(2t_g+5)}{18} \qquad (11)$$

where t_g is the size of the g-th group of tied observations and the standardized test statistic is given by:

$$Z=\begin{cases} \frac{S-1}{\sqrt{Var\ (S)}}, & S>0, \\ 0, & S=0, \\ \frac{S+1}{\sqrt{Var\ (S)}}, & S<0. \end{cases} \qquad (12)$$

It may also be useful to report Kendall's τ coefficient as a measure of the direction and strength of the ordinal association. The standardized Mann–Kendall Z statistic follows an approximately standard normal distribution under the null hypothesis of independent and identically distributed observations. In the absence of ties, Kendall's τ is computed as follows:

$$\tau=\frac{S}{\binom{T}{2}}=\frac{2S}{T(T-1)} \qquad (13)$$

A complementary analysis for the numerical intensity of the trend pattern may be provided by Sen's slope [see Sen (1968)] that is based on the slopes between all the pairs of observations:

$$Q_{ij}=\frac{\hat{\alpha}_j-\hat{\alpha}_i}{j-i},\ j>i \qquad (14)$$

where the estimator is given by the median of those slopes as $\hat{\beta}_{Sen}=median\ (Q_{ij})$.

For the annual top-100 MLE tail exponents over 1999–2024 (T = 26), the Mann–Kendall test yields S = −235 and a continuity-corrected standardized statistic Z = −5.158, with a two-sided normal-approximation p-value of 2.500×10^{-7}. Kendall's τ = −0.723 indicates a negative

ordinal association with time, and Sen's slope is −0.014 per year. The mean Hill exponent is 1.267 in 1999–2009 and 1.050 in 2010–2024; several estimates in 2018–2023 are at or below 1, with a rebound to 1.119 in 2024. These estimates reveal a persistent downward shift between the pre-2010 and post-2010 periods, consistent with the structural break identified below. The Mann–Kendall calibration assumes independent annual observations; its p-value should therefore be interpreted conditionally on that assumption.

Figure 1 motivates a formal assessment of a structural break in the annual Hill exponents. Following the supremum testing approach of Andrews (1993), we use a weighted sup-Wald mean-shift test. For each candidate split, separate segment means are estimated using inverse-variance weights. The maximum Wald statistic tests a constant mean against a single level shift:

$$\text{Sup}\, W = \sup_{\pi_1 \le \pi \le \pi_2} W_T(\pi) \quad (15)$$

where $W_T(\pi)$ denotes the Wald statistic for a candidate split at fraction π of the T annual observations. The search interval excludes endpoint splits.

The break search excludes candidate splits near the sample endpoints. With T = 26, the original weighted sup-Wald calculation searches splits k = 4,…,21, where k is the number of annual observations in the first segment. The maximum statistic is 22.105 and selects the split after 2009. A heteroskedastic parametric bootstrap under the no-break null, repeating the same candidate-date search in B = 20,000 replications, produces two exceedances; the corrected bootstrap p-value is (2 + 1)/(20,000 + 1) = 0.000150. The test therefore rejects the null of no structural break and identifies 2010 as the first year of the new regime. This calibration treats annual estimation errors as independent. We also report Pettitt's (1979) rank-based change-point test, which is given by:[7]

$$U_{t,T} = \sum_{i=1}^{t} \sum_{j=t+1}^{T} sgn\ (\hat{\alpha}_i - \hat{\alpha}_j) \quad (16)$$

[7] With T = 26, the weighted sup-Wald statistic is calibrated by a heteroskedastic parametric bootstrap. The original Python routine searches k = 4,…,21, weights the annual Hill estimates by the reciprocal of their squared standard errors, and retains the maximum mean-shift Wald statistic 22.105. Under the no-break null, each simulated annual estimate equals the common weighted mean plus an independent normal error with variance equal to that year's squared standard error. The full search is repeated in B = 20,000 replications. Two simulated maxima exceed the observed maximum, giving the corrected p-value (2 + 1)/(20,000 + 1) = 0.000150. The maximizing split is after 2009, with weighted means 1.265 before and 1.045 after the split. The search admits at least four observations in the first segment and five in the second. Pettitt's test also selects the split after 2009 (K = 165, approximate p = 0.000260).

The estimated change point is

$$\hat{\tau} = arg\,max_{1 \le t < T} \left| U_{t,T} \right| \qquad (17)$$

Under the null hypothesis of no structural break and assuming independent annual observations, the approximate p-value is given by:

$$p \approx 2 \exp\left[\frac{-6K^2}{T^3 + T^2} \right] \text{where } K = \underset{1 \le t < T}{max} \left| U_{t,T} \right|. \quad (18)$$

In the present application, K = 165 and the approximate p-value is 0.000260. Pettitt's test rejects the null of no structural break and corroborates the weighted sup-Wald result. Both tests locate the break between 2009 and 2010, identifying 2010 as the start of the post-break regime. The agreement between the two tests supports the evidence of a structural change in the upper-tail exponent. The result is interesting as it is consistent with the commodity boom that would reach a peak in 2011. In the Brazilian case, massive subsidized funding was provided by the Brazilian development bank (Banco Nacional de Desenvolvimento Econômico e Social-BNDES) under the industrial policy strategy known as the national champions' policy, which aimed at targeting specific firms in strategic sectors to make them global giants able to compete in the international market. Such policy mainly took place during 2007–2013. An emblematic example is given by JBS that from a relatively smaller scale became the largest firm of the world in the sector of animal protein, especially after the acquisition of major international competitors (Swift and Pilgrim's Pride) after the favorable environment following the commodities' boom. Such industrial policy also benefited firms outside the commodities' segment as in telecommunications, technology and infrastructure. The adequacy of such policy is controversial but clearly the rise of superstar firms has a role in determining the thickness of the upper tail of the firm-size distribution. [8]

The detected structural break in 2010 reinforces the evidence of a thicker upper tail of the firm size distribution. The persistence of the post-break pattern can, in principle, be consistent with important heterogeneities of the Brazilian economy. It is possible that economic mechanisms associated with agglomeration and a propagation mechanism like co-agglomeration might be having a role in generating the referred power law pattern only for the upper tail of the distribution. In the case of the Brazilian manufacturing industry, there are some

[8] Autor et al. (2020) highlight the decline associated with labor share in the context of the rising of superstar firms that would attempt to focus on productivity improvements. In fact, the explosive growth of artificial intelligence (AI) utilization may reflect that tendency. The scenario with giant firms may indicate more consolidated conditions for a thicker upper tail of the firm size distributions.

studies in Brazil that build on the spatial concentration measurement approaches advanced by Devereux et al. (2004), Ellison and Glaeser (1997, 1999) and Ellison et al. (2010) and Maurel and Sédillot (1999).

The empirical restriction of power law validity mostly for the 100 largest firms in Brazil requires an examination beyond purely stochastic generative mechanisms. This distinct statistical behavior in the extreme upper tail can be contextualized in terms of empirical economic geography in emerging markets. Specifically, the localized nature of industrial concentration in Brazil, characterized by severe sectoral heterogeneity and infrastructure constraint, suggests that Marshallian externalities do not scale linearly across the firm size distribution. Almeida and Rocha (2018) found econometric evidence of labor pooling in terms of sharing a specialized workforce in the Brazilian Northeast from 2002 to 2014, and indicated its positive role in explaining regional geographic concentration.

Resende and Wyllie (2005) undertook a comprehensive country-wide study on industrial agglomeration upon detailed firm-level data for Brazil over the 1995–2001 period. The evidence indicated that spatial agglomeration in Brazilian manufacturing is heavily concentrated within highly specialized hubs. It is likely that only the elite tier of the largest 100 firms possesses the localized infrastructure layout necessary to internalize these external economies of scale.

A possible relevant propagation mechanism for the aforementioned agglomeration effects, pertains to co-agglomeration when the structural dynamics of industrial agglomeration introduce interactive multi-sector multipliers that reinforce this tail extremity. Resende (2015) investigates the case of the state of Rio de Janeiro-Brazil in 2010 and the evidence indicated that the pairing of manufacturing sectors in space is heavily driven by labor pooling and intensive input-output linkages.

Altogether, the distinctive pattern associated with the upper tail of firm size distribution seems consistent with structural features of the Brazilian economy but one needs to be cautious and not force causal explanations. Clearly, one needs to advance in econometric analyses to uncover the underlying economic explanatory factors that provide the basis for the prevalence of power laws.

4. Final comments

The paper has investigated distributional properties of the largest firms in Brazil during the 1999–2024 period, taking as a reference net revenue shares. The main focus of the analysis refers to the assessment of the prevalence of simple power law patterns where a single shape parameter properly portrays the firm size distribution for its upper tail (1000, 500 and 100

largest firms). Specifically, the Pareto type I case or its particular case of the Zipf law were nested either within a Pareto type II or a more general Pareto type IV model.

The evidence, thus obtained for a power law, is only stronger for the upper tail of the distribution corresponding to the largest 100 firms. In that portion, the case for a Pareto type model and even a Zipf model is appealing for a significant portion of the studied years. However, the power-law hypothesis is systematically rejected under a less strict cutoff point (500 and 1000 largest firms) in virtually all years.

The annual Pareto exponents for the largest 100 firms reveal a statistically significant downward trend and a structural break in 2010. The Mann–Kendall test detects the declining trend, while the weighted sup-Wald and Pettitt tests reject the null of no structural break and identify the split after 2009. The mean Hill exponent falls from 1.267 in 1999–2009 to 1.050 in 2010–2024, signaling a thicker upper tail and rising revenue concentration among superstar firms. The timing of the break is consistent with the commodity-boom period and motivates further investigation of its economic drivers.

Our findings suggest several avenues for future research. One promising direction is to investigate the drivers of the structural break in 2010, exploring how shifts in credit policy (such as lending from the Brazilian development bank BNDES), tax regimes, or trade shocks affected the growth of the largest firms, but foremost it is possible to detect two important gaps in the literature in general and of that pertaining to the Brazilian case:

i) An econometric investigation on the underlying economic factors explaining the Pareto exponent. In fact, possible consistencies with industrial agglomeration and coagglomeration patterns have been highlighted but the more primitive associated economic factors could be considered in an econometric study to enable causal assessments;

ii) Investigation of the propagation of microeconomic shocks to macroeconomic shocks in terms of analyses of granularity that can emerge in the context of a very thick upper tail of the firm size distribution. The related literature for Brazil is still scarce. Da Silva et al. (2018) attempt to make statements on granularity solely on some evidence for power law prevalence for a given year with the upfront assumption of particular distributions with the absence of detailed testing and do not provide a quantification of the so-called granular residual and its possible aggregate impact. Santos (2026), on the other hand, builds on Gabaix (2011) and Blanco-Arroyo et al. (2018) and makes an initial effort on that neglected aspect. The study considers data sets of the largest 1000 firms in Brazil in terms of net revenues, over the 2000–2019 period and investigated the granular hypothesis. The quantification of the granular residual required matching of the database with employment data from another database so as to compute a proxy for labor productivity growth necessary for the computation. The evidence indicated sizable indicators that were considered in terms of a dynamic ARDL model to assess

the aggregate impact on GDP. The econometric effort is limited by the small sample size but clearly such line of research appears to show unexplored potential for future investigations.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Table 1

Lagrange multiplier tests (LM) for restricted Pareto type II distributions for the 1000 largest firms in Brazil – 1999–2024

Year	LMp [H_0: $\sigma = \mu$]			LMz [H_0: $\sigma = \mu$, $\alpha = 1$]		
	test statistic	asy. p-value	boot. p-value	test statistic	asy. p-value	boot. p-value
1999	27.870	<0.001	0.001	64.895	<0.001	0.001
2000	31.541	<0.001	0.001	51.325	<0.001	0.001
2001	24.685	<0.001	0.001	35.554	<0.001	0.001
2002	34.206	<0.001	0.001	55.244	<0.001	0.001
2003	23.200	<0.001	0.001	44.346	<0.001	0.001
2004	15.632	<0.001	0.001	33.384	<0.001	0.001
2005	8.084	0.004	0.002	21.541	<0.001	0.001
2006	10.534	0.001	0.003	20.956	<0.001	0.001
2007	12.745	<0.001	0.001	27.074	<0.001	0.001
2008	7.165	0.007	0.008	16.866	<0.001	0.001
2009	10.305	0.001	0.005	20.231	<0.001	0.001
2010	20.807	<0.001	0.001	67.855	<0.001	0.001
2011	15.194	<0.001	0.001	59.068	<0.001	0.001
2012	18.589	<0.001	0.001	52.522	<0.001	0.001
2013	23.962	<0.001	0.001	71.820	<0.001	0.001
2014	29.130	<0.001	0.001	79.676	<0.001	0.001
2015	31.692	<0.001	0.001	88.075	<0.001	0.001
2016	20.896	<0.001	0.001	83.457	<0.001	0.001
2017	54.399	<0.001	0.001	151.423	<0.001	0.001
2018	30.702	<0.001	0.001	111.622	<0.001	0.001
2019	31.911	<0.001	0.001	120.286	<0.001	0.001
2020	50.135	<0.001	0.001	182.549	<0.001	0.001
2021	32.597	<0.001	0.001	129.970	<0.001	0.001
2022	24.621	<0.001	0.001	109.477	<0.001	0.001
2023	33.149	<0.001	0.001	118.653	<0.001	0.001
2024	38.099	<0.001	0.001	127.200	<0.001	0.001

Notes: LMp tests H_0: $\sigma = \mu$ (Pareto type I within Pareto type II, $\chi^2(1)$); LMz tests H_0: $\sigma = \mu$ and $\alpha = 1$ (Zipf within Pareto type II, $\chi^2(2)$). Both tests use the smallest included revenue as the cutoff. Asymptotic probabilities are chi-squared upper-tail probabilities. Bootstrap probabilities use 999 parametric replications and the correction $(k + 1)/(999 + 1)$, where k is the number of simulated statistics at least as large as the observed statistic. The retained historical Python calculation uses seed 20260701 for LMp and a separate stream with seed 20260702 for LMz; the exponent under the LMp null is the restricted estimate, while LMz imposes $\alpha = 1$. The simulated sample minimum is re-estimated in every replication. Recomputed bootstrap values in the V12 main_tests.csv use seed 20260923 and constitute a separate Monte Carlo run. The 2019 top-1000 sample includes the restored rank-779 firm in the V12 analysis; its displayed statistics were updated, and its bootstrap probabilities remain 0.001 at the displayed precision.

Table 2

Lagrange multiplier tests (LM) for restricted Pareto type II distributions for the 500 largest firms in Brazil – 1999–2024

Year	LMp [H_0: $\sigma = \mu$]			LMz [H_0: $\sigma = \mu$, $\alpha = 1$]		
	test statistic	asy. p-value	boot. p-value	test statistic	asy. p-value	boot. p-value
1999	8.045	0.005	0.004	8.036	0.018	0.014
2000	10.619	0.001	0.002	11.082	0.004	0.003
2001	10.957	0.001	0.003	10.968	0.004	0.006
2002	12.934	<0.001	0.001	12.759	0.002	0.004
2003	12.134	<0.001	0.003	12.712	0.002	0.004
2004	6.668	0.010	0.007	6.810	0.033	0.028
2005	8.111	0.004	0.005	8.305	0.016	0.015
2006	10.090	0.001	0.003	10.887	0.004	0.013
2007	9.601	0.002	0.003	10.879	0.004	0.007
2008	8.467	0.004	0.003	9.564	0.008	0.012
2009	8.227	0.004	0.004	8.271	0.016	0.020
2010	6.950	0.008	0.014	12.185	0.002	0.002
2011	3.892	0.049	0.051	7.978	0.019	0.019
2012	0.979	0.322	0.327	1.524	0.467	0.463
2013	3.705	0.054	0.053	5.292	0.071	0.078
2014	2.213	0.137	0.137	2.866	0.239	0.270
2015	3.032	0.082	0.075	4.503	0.105	0.097
2016	9.741	0.002	0.004	14.271	0.001	0.001
2017	8.703	0.003	0.005	12.725	0.002	0.003
2018	5.332	0.021	0.022	9.189	0.010	0.011
2019	5.712	0.017	0.013	9.859	0.007	0.007
2020	5.255	0.022	0.036	19.571	<0.001	0.001
2021	7.771	0.005	0.006	17.298	<0.001	0.001
2022	8.410	0.004	0.006	18.141	<0.001	0.001
2023	4.105	0.043	0.051	8.832	0.012	0.010
2024	5.747	0.017	0.014	12.902	0.002	0.003

Notes: LMp tests H_0: $\sigma = \mu$ (Pareto type I within Pareto type II, $\chi^2(1)$); LMz tests H_0: $\sigma = \mu$ and $\alpha = 1$ (Zipf within Pareto type II, $\chi^2(2)$). Both tests use the smallest included revenue as the cutoff. Asymptotic probabilities are chi-squared upper-tail probabilities. Bootstrap probabilities use 999 parametric replications and the correction $(k + 1)/(999 + 1)$, where k is the number of simulated statistics at least as large as the observed statistic. The retained historical Python calculation uses seed 20260701 for LMp and a separate stream with seed 20260702 for LMz; the exponent under the LMp null is the restricted estimate, while LMz imposes $\alpha = 1$. The simulated sample minimum is re-estimated in every replication. Recomputed bootstrap values in the V12 main_tests.csv use seed 20260923 and constitute a separate Monte Carlo run.

Table 3

Lagrange multiplier tests (LM) for restricted Pareto type II distributions for the 100 largest firms in Brazil – 1999–2024

Year	LMp [H_0: $\sigma = \mu$]			LMz [H_0: $\sigma = \mu$, $\alpha = 1$]		
	test statistic	asy. p-value	boot. p-value	test statistic	asy. p-value	boot. p-value
1999	0.247	0.619	0.636	5.088	0.079	0.098
2000	0.254	0.615	0.611	7.079	0.029	0.031
2001	1.157	0.282	0.262	5.051	0.080	0.086
2002	1.086	0.297	0.314	4.704	0.095	0.092
2003	1.424	0.233	0.254	4.541	0.103	0.110
2004	2.094	0.148	0.150	4.679	0.096	0.104
2005	1.362	0.243	0.234	4.018	0.134	0.124
2006	1.274	0.259	0.262	4.887	0.087	0.084
2007	2.521	0.112	0.121	4.246	0.120	0.102
2008	0.534	0.465	0.475	4.437	0.109	0.092
2009	1.345	0.246	0.241	4.103	0.129	0.120
2010	2.158	0.142	0.107	2.185	0.335	0.316
2011	0.456	0.500	0.481	1.252	0.535	0.539
2012	0.610	0.435	0.419	1.137	0.566	0.579
2013	0.797	0.372	0.349	1.433	0.488	0.491
2014	1.299	0.254	0.229	1.774	0.412	0.418
2015	0.598	0.439	0.449	1.460	0.482	0.503
2016	5.229	0.022	0.022	5.027	0.081	0.085
2017	5.289	0.021	0.028	5.099	0.078	0.087
2018	8.725	0.003	0.005	8.825	0.012	0.017
2019	8.861	0.003	0.003	9.411	0.009	0.010
2020	4.627	0.031	0.032	4.485	0.106	0.106
2021	3.001	0.083	0.071	2.953	0.228	0.232
2022	6.191	0.013	0.009	6.498	0.039	0.035
2023	4.885	0.027	0.019	4.906	0.086	0.088
2024	1.164	0.281	0.265	1.925	0.382	0.400

Notes: LMp tests H_0: $\sigma = \mu$ (Pareto type I within Pareto type II, $\chi^2(1)$); LMz tests H_0: $\sigma = \mu$ and $\alpha = 1$ (Zipf within Pareto type II, $\chi^2(2)$). Both tests use the smallest included revenue as the cutoff. Asymptotic probabilities are chi-squared upper-tail probabilities. Bootstrap probabilities use 999 parametric replications and the correction $(k + 1)/(999 + 1)$, where k is the number of simulated statistics at least as large as the observed statistic. The retained historical Python calculation uses seed 20260701 for LMp and a separate stream with seed 20260702 for LMz; the exponent under the LMp null is the restricted estimate, while LMz imposes $\alpha = 1$. The simulated sample minimum is re-estimated in every replication. Recomputed bootstrap values in the V12 main_tests.csv use seed 20260923 and constitute a separate Monte Carlo run.

Table 4

Cutoff sensitivity of exact Pareto type I tests across 26 annual rankings

Top n	Years	LMp rejects	Mean α	Min α	Max α
50	26	2	1.311	1.065	1.589
75	26	8	1.172	1.061	1.272
100	26	9	1.129	0.940	1.325
125	26	5	1.126	0.950	1.321
150	26	2	1.120	0.975	1.284
200	26	2	1.110	0.993	1.297
300	26	10	1.035	0.823	1.216
500	26	23	0.948	0.848	1.058
750	26	26	0.884	0.818	0.980
999	26	26	0.823	0.715	0.907

Notes: The sample comprises the top n firms, with μ set to the revenue of rank n+1. Rejection counts use asymptotic LMp $p < 0.05$; they are descriptive across a prespecified grid, not selection-adjusted test decisions. The n = 999 row uses the first 999 firms and rank 1000 as cutoff.

Table 5

Lagrange multiplier tests (LM) for restricted Pareto type IV distributions for the 100 largest firms in Brazil – 1999–2024

Year	PWL [H_0: $\sigma = \mu$, $\gamma = 1$]		
	test statistic	asy. p-value	boot. p-value
1999	0.564	0.754	0.752
2000	3.888	0.143	0.134
2001	5.650	0.059	0.054
2002	2.466	0.291	0.285
2003	3.500	0.174	0.175
2004	3.173	0.205	0.198
2005	6.731	0.035	0.034
2006	2.185	0.335	0.332
2007	4.397	0.111	0.105
2008	4.235	0.120	0.116
2009	2.639	0.267	0.258
2010	3.591	0.166	0.165
2011	1.412	0.494	0.486
2012	1.767	0.413	0.409
2013	2.821	0.244	0.236
2014	1.385	0.500	0.499
2015	0.981	0.612	0.610
2016	5.931	0.052	0.049
2017	5.850	0.054	0.052
2018	9.130	0.010	0.011
2019	11.131	0.004	0.006
2020	5.407	0.067	0.064
2021	3.963	0.138	0.134
2022	6.298	0.043	0.043
2023	5.528	0.063	0.060
2024	3.249	0.197	0.187

Notes: PWL tests H_0: $\sigma = \mu$ and $\gamma = 1$ against Pareto type IV (χ^2 with two degrees of freedom). μ is the revenue of the first excluded firm, rank 101, fixed within each annual conditional-tail bootstrap. The bootstrap p-values use 19,999 replications, with an independent fixed seed for every fiscal year.

Table 6

Estimated tail exponents and tests of Zipf's law for the top 100 firms in 1999–2024

Year	MLE $\hat{\alpha}$	MLE asy. p-value	MLE boot. p-value	OLS-GI $\hat{\alpha}$	OLS-GI asy. p-value	OLS-GI boot. p-value	MLE absolute z
1999	1.291	0.024	0.025	1.352	0.066	0.069	2.254
2000	1.359	0.008	0.010	1.444	0.030	0.037	2.644
2001	1.283	0.027	0.028	1.424	0.035	0.041	2.208
2002	1.270	0.034	0.033	1.402	0.042	0.047	2.125
2003	1.257	0.041	0.039	1.404	0.042	0.047	2.043
2004	1.248	0.047	0.046	1.415	0.038	0.044	1.990
2005	1.234	0.058	0.059	1.375	0.054	0.058	1.893
2006	1.275	0.031	0.031	1.405	0.042	0.047	2.155
2007	1.212	0.080	0.080	1.388	0.048	0.053	1.748
2008	1.265	0.036	0.035	1.353	0.065	0.068	2.098
2009	1.238	0.054	0.055	1.368	0.057	0.061	1.923
2010	1.049	0.641	0.643	1.208	0.224	0.216	0.467
2011	1.109	0.327	0.324	1.201	0.236	0.226	0.980
2012	1.090	0.409	0.408	1.188	0.264	0.253	0.826
2013	1.102	0.353	0.352	1.209	0.221	0.214	0.928
2014	1.097	0.378	0.376	1.211	0.219	0.211	0.881
2015	1.116	0.299	0.296	1.205	0.229	0.220	1.038
2016	1.053	0.616	0.619	1.257	0.148	0.143	0.502
2017	1.043	0.681	0.682	1.260	0.144	0.140	0.411
2018	0.992	0.933	0.935	1.264	0.140	0.137	0.084
2019	0.961	0.686	0.688	1.240	0.171	0.165	0.404
2020	1.037	0.720	0.720	1.248	0.160	0.154	0.358
2021	1.017	0.867	0.867	1.183	0.274	0.262	0.168
2022	0.970	0.758	0.759	1.189	0.261	0.249	0.308
2023	0.997	0.975	0.976	1.210	0.219	0.212	0.031
2024	1.119	0.289	0.285	1.247	0.161	0.155	1.060

Notes: All p-values test H_0: $\alpha = 1$ against H_1: $\alpha \neq 1$. MLE is the Pareto type I maximum-likelihood (Hill) estimator with the smallest included revenue as the cutoff. OLS-GI is the Gabaix–Ibragimov rank-size estimator. Asymptotic p-values are two-sided normal probabilities based on $z = (\hat{\alpha} - 1)/\mathrm{SE}$, with $\mathrm{SE} = \hat{\alpha}/\sqrt{100}$ for MLE and $\mathrm{SE} = \hat{\alpha}\sqrt{(2/100)}$ for OLS-GI. MLE bootstrap p-values use 20,000 parametric draws under $\alpha = 1$, the fixed-cutoff simulation convention of the retained calculation, and the correction $(k + 1)/(20{,}000 + 1)$. The same null simulation draws are used to calibrate all annual MLE statistics. OLS-GI bootstrap p-values use the same 20,000 parametric draws under $\alpha = 1$, with the rank-size exponent re-estimated in each draw and the same $(k + 1)/(20{,}000 + 1)$ correction. Standard errors are used internally for inference but are omitted from the table. Three decimal places report the recorded values; bootstrap simulation uncertainty remains.

Figure 1

Estimated Pareto exponents for the largest 100 firms in Brazil – 1999–2024

